Moments Continued

Relations Between Raw Moments And Central Moments

As previously discussed, **Raw Moments (μ'_r)** are calculated about an arbitrary point 'A' (often the origin, $A = 0$), while **Central Moments (μ_r)** are calculated about the **mean ($\bar{x}$)**.

Raw moments are often easier to calculate. We can then convert them into the more descriptive central moments using a set of standard relationships.

Let $\mu'_1, \mu'_2, \mu'_3, \mu'_4$ be the first four raw moments calculated about an arbitrary point 'A'.
Let $\bar{x}$ be the true mean of the data.

First, we establish the distance between the arbitrary point 'A' and the true mean $\bar{x}$.
Let $d = \bar{x} - A$.
We know that $\bar{x} = A + \mu'_1$ (from the definition of μ'_1), so $d = \mu'_1$.

The relationships to convert the first four raw moments to central moments are:

- **First Central Moment (μ_1):**

$$\mu_1 = 0$$

  This is true by definition. The sum of deviations from the mean is always zero.

- **Second Central Moment (μ_2):**

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

  This is the most famous relationship. It states: **Variance = (Second raw moment) - (First raw moment squared)**.

- **Third Central Moment (μ_3):**

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3$$

- **Fourth Central Moment (μ_4):**

$$\mu_4 = \mu'_4 - 4\mu'_1\mu'_3 + 6(\mu'_1)^2\mu'_2 - 3(\mu'_1)^4$$

These formulas are the tools that allow us to use a simpler calculation method (raw moments) to find the more meaningful values (central moments) used to describe variance, skewness, and kurtosis.

Skewness

Definition: Skewness is a measure of the **asymmetry** or **lopsidedness** of a probability distribution. A perfectly symmetrical distribution (like a bell curve) has zero skewness.

- **Symmetrical Distribution (Zero Skewness):** The data values are evenly distributed on both sides of the center.
  - **Mean = Median = Mode**
- **Positively Skewed Distribution (Skewed Right):**
  - The "tail" of the distribution is longer on the **right side**.
  - This means there are a few unusually **high** values that pull the mean to the right.
  - **Mean > Median > Mode**
- **Negatively Skewed Distribution (Skewed Left):**
  - The "tail" of the distribution is longer on the **left side**.
  - This means there are a few unusually **low** values that pull the mean to the left.
  - **Mode > Median > Mean**

Skewness tells us the direction and relative magnitude of a distribution's deviation from symmetry.

Karl Pearson's Coefficient of skewnesss

This is a **relative measure** of skewness, meaning it gives a single number that can be compared across different datasets (regardless of their units). It is based on the relationship between the mean, median, and mode.

Formula 1 (Based on Mode):

$$= \frac{\text{Mean} - \text{Mode}}{\text{Standard eiation}}$$

- **Interpretation:**
  - If $\text{Mean} > \text{Mode}$, the coefficient is positive (positively skewed).
  - If $\text{Mean} < \text{Mode}$, the coefficient is negative (negatively skewed).
  - If $\text{Mean} = \text{Mode}$, the coefficient is 0 (symmetrical).

Formula 2 (Based on Median):
This formula is preferred when the mode is ill-defined (e.g., multimodal data). It uses the empirical relationship: $\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}$.

$$= \frac{3(\text{Mean} - \text{Median})}{\text{Standard eiation}}$$

- **Note:** The value of Pearson's coefficient typically lies between **-3 and +3**. For most distributions, it lies between -1 and +1.

Bowley's Method

Bowley's coefficient of skewness is also a relative measure, but it is based on the **quartiles** (Q_1, Q_2, Q_3). It measures skewness based on the middle 50% of the data, which makes it **robust to outliers** (not affected by extreme values).

Definition: It compares the distance of the third quartile (Q_3) from the median (Q_2) with the distance of the median (Q_2) from the first quartile (Q_1).

- In a symmetrical distribution, $\text{Median} - Q_1 = Q_3 - \text{Median}$.
- In a positively skewed distribution, $Q_3 - \text{Median} > \text{Median} - Q_1$.
- In a negatively skewed distribution, $Q_3 - \text{Median} < \text{Median} - Q_1$.

Formula:

$$= \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{Q_3 - Q_1}$$

This simplifies to:

$$= \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

(Where Q_2 is the Median)

- **Interpretation:** The value of Bowley's coefficient always lies between **-1 and +1**.
  - > 0: Positively skewed
  - < 0: Negatively skewed
  - $= 0$: Symmetrical

Method Of Moments

This is the most mathematically precise method for measuring skewness. It uses the **third central moment (μ_3)**, which is inherently a measure of asymmetry.

We know $\mu_3 = \frac{\sum f_i (x_i - \bar{x})^3}{N}$.

- If the distribution is symmetrical, the positive and negative deviations $(x_i - \bar{x})^3$ will cancel each other out, and $\mu_3 = 0$.
- If the distribution is positively skewed, the large positive deviations (cubed) will overpower the negative ones, and $\mu_3 > 0$.
- If the distribution is negatively skewed, the large negative deviations (cubed) will overpower the positive ones, and $\mu_3 < 0$.

However, μ_3 is an **absolute measure**. Its value depends on the units of the data (e.g., cm^3). To make it a **relative measure** that is unit-free, we standardize it by dividing by the cube of the standard deviation (σ^3). This leads to the coefficient of skewness based on moments.

Kurtosis

Definition: Kurtosis measures the **"peakedness"** or **"tailedness"** of a distribution compared to a **Normal Distribution** (the standard bell curve). It describes how concentrated the data is around the mean and how heavy the tails are.

Kurtosis is measured by the **fourth central moment (μ_4)**.

We classify distributions into three types based on their kurtosis:

1. **Mesokurtic:**
   - This is the benchmark. The distribution has the same peakedness and tail weight as a **Normal Distribution**.
   - Kurtosis value (see coefficient below) is 3.
2. **Leptokurtic:**
   - The distribution is **more peaked** (lepto = "slender") than a normal distribution.
   - It has a sharper peak and **heavier tails**. This means there is a higher concentration of values near the mean, *and* a higher probability of extreme outlier values.
   - Kurtosis value is > 3.
3. **Platykurtic:**
   - The distribution is **flatter** (platy = "broad") than a normal distribution.
   - It has a lower, wider peak and **lighter tails**. This means there is a lower concentration of values near the mean and a lower probability of extreme values.
   - Kurtosis value is < 3.

Coefficient Of Skewness and Kurtosis Based On Moments

These are the unit-free, relative measures of skewness and kurtosis derived from the central moments. They are often denoted by the Greek letters Beta (β) and Gamma (γ).

Coefficient of Skewness (β_1 and γ_1):

This coefficient is derived from the third central moment (μ_3) and the second central moment (μ_2, which is variance).

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

- β_1 is always positive (due to squaring μ_3), so it only measures the *magnitude* of skewness, not the direction.
- For a symmetrical distribution, $\mu_3 = 0$, so $\beta_1 = 0$.

To get the direction (positive or negative), we use γ_1 (gamma one), which is the square root of β_1:

$$\gamma_1 = \sqrt{\beta_1} = \frac{\mu_3}{\sqrt{\mu_2^3}} = \frac{\mu_3}{\sigma^3}$$

(The sign of γ_1 is taken from the sign of μ_3)

- **Interpretation (γ_1):**
  - $\gamma_1 > 0$: Positively skewed
  - $\gamma_1 < 0$: Negatively skewed
  - $\gamma_1 = 0$: Symmetrical

Coefficient of Kurtosis (β_2 and γ_2):

This coefficient is derived from the fourth central moment (μ_4) and the second central moment (μ_2).

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{\sigma^4}$$

- This is the primary measure of kurtosis.
- **Interpretation (β_2):**
  - $\beta_2 = 3$: **Mesokurtic** (Normal distribution)
  - $\beta_2 > 3$: **Leptokurtic** (Peaked, heavy tails)
  - $\beta_2 < 3$: **Platykurtic** (Flat, light tails)

Excess Kurtosis (γ_2):

Often, statisticians prefer to measure kurtosis *relative to* the normal distribution (which has $\beta_2 = 3$). This is called "excess kurtosis."

$$\gamma_2 = \beta_2 - 3$$

- **Interpretation (γ_2):**
  - $\gamma_2 = 0$: **Mesokurtic**
  - $\gamma_2 > 0$: **Leptokurtic**
  - $\gamma_2 < 0$: **Platykurtic**
  - This is why you will often see "Kurtosis = 0" as the benchmark; it is referring to the *excess kurtosis*.